Connors 3-Day Hi-Lo Strategy

Backtest, Analysis & Notebook

S&P 500 | 2003–2025 | 22-Year Analysis

A comprehensive guide to the Connors 3-Day Hi-Lo mean-reversion strategy — covering entry logic, pattern recognition, risk metrics, Monte Carlo simulation, and professional execution insights.

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Introduction to the Connors 3-Day Hi-Lo Strategy

What is the Connors 3-Day Hi-Lo Strategy?

The Connors 3-Day Hi-Lo is a short-term mean-reversion strategy developed by Larry Connors and Cesar Alvarez. It identifies brief, structured pullbacks within an established uptrend by requiring three consecutive days of sequentially lower highs AND lower lows — a precise, multi-bar compression pattern that signals exhausted selling pressure.

Unlike single-bar oversold indicators (such as RSI or stochastics), this strategy uses a pure price-action pattern: three days of step-by-step price compression. The logic is that when the market carves out three consecutive lower-high, lower-low candles while still above its long-term average, it has temporarily detached from its upward trajectory and is primed to snap back.

The Core Concept

Condition	Action
Close ABOVE 200-day SMA AND BELOW 5-day SMA AND 3 consecutive lower highs & lower lows	BUY — Compressed pullback in confirmed uptrend
Close CROSSES ABOVE 5-day SMA	EXIT — Short-term momentum has recovered

Think of the 200-day SMA as the long-term regime filter (only buy in uptrends) and the 5-day SMA as the short-term mean (the rubber band). The strategy fires when price has been compressed below the 5-day SMA for three structured days, then profits when it snaps back above it.

Why Three Consecutive Lower Highs AND Lower Lows?

- **Precision filter:** A single down day is noise. Two down days may be coincidence. Three consecutive days of sequentially declining highs AND lows is a structured, orderly pullback — not a crash.
- **Measured compression:** Each new lower high confirms sellers are in control short-term; each new lower low confirms the range is compressing. This compression creates the “coiled spring” for the subsequent rebound.
- **Noise reduction:** Requiring BOTH a lower high AND a lower low on each of the three days eliminates inside bars, doji candles, and other ambiguous price action that single-indicator strategies would misidentify as entry signals.

The Dual SMA Architecture

Filter	Parameter	Purpose
Long-term trend filter	200-day SMA	Confirms bullish market regime. Prevents buying into a falling market where mean-reversion can continue much lower.
Short-term mean filter	5-day SMA	Acts as the “Rubber Band.” Entry fires when price is below it (stretched); exit fires when price recrosses above it (snapped back).

Entry Logic: When do we ENTER a trade?

Step	Condition
1	Confirm close is ABOVE the 200-day SMA (uptrend regime active)

2	Confirm close is BELOW the 5-day SMA (short-term weakness present)
3	Confirm 2 days ago: both high AND low were LOWER than the day before that
4	Confirm yesterday: both high AND low were LOWER than 2 days ago
5	Confirm today: both high AND low are LOWER than yesterday
6	ALL conditions met → Enter LONG position at today's close (or next open)
7	Any condition fails → No trade; remain in cash

The pattern requires simultaneous lower highs AND lower lows on each day — not just a lower close. This rules out days where price opened high (lower close but higher high) or closed weak (lower high but lower low via gap). It is a very precise, symmetrical compression.

Exit Logic: When do we EXIT a trade?

Step	Condition
1	Monitor position daily after entry
2	Check if close has crossed ABOVE the 5-day SMA
3	If YES → Exit position at close (or next open)
4	If NO → Hold position; the short-term mean-reversion has not yet completed

The 5-day SMA exit is elegant: it is the same reference point used for entry (the short-term mean), creating a symmetrical trade structure. You enter when price is below the rubber band, and exit when it has snapped back above it.

Key Definitions

Term	Definition
Lower High	Today's intraday high is below yesterday's intraday high — sellers capped the rally at a lower point
Lower Low	Today's intraday low is below yesterday's intraday low — buyers failed to defend the prior floor
3-Day Compression	Three consecutive days with BOTH lower highs AND lower lows — an orderly, structured pullback
5-day SMA	The average closing price of the last 5 days — acts as the short-term mean / rubber band
200-day SMA	The long-term trend filter — only trade long when price is above this level
Mean Reversion	The tendency of compressed prices to snap back toward their recent average
Drawdown	The percentage decline from a peak account value to a subsequent trough
Whipsaw	Entry fires but price continues lower rather than reverting, resulting in a loss

How This Strategy Works

1. Pattern-Based Signal Quality

By requiring three specific days of structured compression — not just an oversold indicator reading — the strategy selects only the highest-quality mean-reversion setups. The pattern is objective, unambiguous, and cannot be faked by a single gap day or volatile bar.

2. The Regime Filter Prevents Catastrophic Losses

The 200-day SMA filter is the strategy's most important component. Without it, three consecutive lower highs and lows could equally describe a market in free-fall. The 200-day SMA ensures signals only fire when the broader trend is up — dramatically improving the probability that the compression is temporary rather than the beginning of a bear market.

3. Ultra-Short Hold Time

- Average Hold: 4.42 days — in and out within a week on average
- Time in Market: only 6.58% — cash idle for over 93% of all trading days
- This extreme brevity minimises overnight gap risk and macroeconomic event exposure

Pros and Cons

Benefit	Cost
High win rate (77%+) — statistically reliable	Average loss is larger than average win
Very short holds — minimal overnight risk	CAGR is modest (2.33%) due to low time in market
Pure price-action pattern — no indicator lag	Requires precise bar-by-bar pattern matching
Excellent Sharpe ratio (2.05)	Signals are infrequent (~5.5 per year)
Near-zero risk of ruin (MC: 0.41%)	Underperforms Buy & Hold during smooth bull runs

Important Warnings

⚠ This is NOT a "Get Rich Quick" Strategy

Expectation	Reality
Large per-trade gains	Small average wins (~1.0%) via high frequency of winning
Beat the market every year	CAGR of 2.33% — lags Buy & Hold in strong bull markets
Frequent signals	~5–6 trades per year — most time spent waiting in cash

This is the trade-off of a high-precision mean-reversion system. It wins frequently but earns modestly per trade. Its primary value is in risk-adjusted returns — the strategy's Sharpe ratio of 2.05 and maximum drawdown of just -13.29% make it one of the most capital-efficient strategies in systematic trading.

SECTION SUMMARY The Connors 3-Day Hi-Lo strategy is a precision pattern-recognition mean-reversion system: wait for three consecutive lower highs AND lower lows while the market remains above its 200-day average, then buy the compressed pullback and exit when the 5-day mean is recrossed. Success requires patience, precise execution, and appreciation for risk-adjusted rather than absolute returns.

Connors 3-Day Hi-Lo Strategy: Buy & Sell Signals (2000-2025)
S&P 500 Close
SMA(200)
SMA(5)
Buy Signal
Sell Signal
Price (USD)
7000
6000
5000
4000
3000
2000
1000
2000
2004
2008
2012
2016
2020
2024

Executive Summary: The “Structured Compression” Edge

Performance Summary: Connors 3-Day Hi-Lo Strategy (2003–2025)

This table consolidates the key performance metrics of the 3-Day Hi-Lo mean-reversion strategy, including profitability, risk, and trade dynamics.

Category / Metric	Value	Professional Interpretation
CAGR	2.33%	Modest absolute returns; exceptional risk-adjusted performance relative to drawdown.
Profit Factor	2.22	Solid. The system returned $2.22 for every $1.00 lost.
Max Drawdown	-13.29%	Outstanding capital preservation across 22+ years of market history.
Annualized Vol	12.16%	Comparable to the S&P 500 due to the high win rate’s variance properties.
Sharpe Ratio	2.05	Institutional quality. Exceptional risk-adjusted return per unit of volatility.
Win Rate	77.05%	High-win, low-payoff system; psychologically very manageable.
Avg Trade PnL	0.44%	Driven by consistency of wins rather than magnitude.
Num Trades	122	Robust sample over 22+ years (~5.5 trades per year).

Comprehensive Performance Metrics

Metric	Value	USD$
Total Net Profit	67.60%	$67,597.67
Gross Profit	97.22%	$97,216.82
Gross Loss	-43.84%	-$43,841.38
Profit Factor	2.22	N/A
Max. Drawdown (Equity)	-13.29%	N/A
Sharpe Ratio	2.05	N/A
Sortino Ratio	1.59	N/A
Ulcer Index	0.0359	N/A
Kelly Criterion	42.30%	N/A
Total # of Trades	122	N/A
Percent Profitable	77.05%	N/A
# of Winning Trades	94	N/A
# of Losing Trades	28	N/A
# of Even Trades	0	N/A
Avg. Trade PnL	0.44%	$436.95
Avg. Winning Trade PnL	1.03%	$1,034.22
Avg. Losing Trade PnL	-1.57%	-$1,565.77
Ratio Avg. Win/Avg. Loss	0.66	N/A
Max. Consec. Winners	10	N/A
Max. Consec. Losers	4	N/A
Largest Winning Trade PnL	3.28%	$3,281.50
Largest Losing Trade PnL	-10.14%	-$10,135.00
Avg. # of Trades per Day	0.015	N/A
Avg. Time in Market	4.42 days	N/A
Profit per Month	0.25%	$251.19
Max. Flat Period	~120 days	N/A
Avg. MAE	-1.50%	N/A
Avg. MFE	1.28%	N/A
Avg. ETD	0.84%	N/A

The Mathematical Engine: Positive Expectancy via Volume and Precision

The 3-Day Hi-Lo achieves positive expectancy through a combination of high win frequency and a pattern filter that selects quality setups. Unlike trend-following systems that win rarely but hugely, this strategy wins 3 out of 4 times at small-but-consistent margins.

The Formula

Expectancy = (Win % × Average Win) − (Loss % × Average Loss)

Component	Value
Win Rate	~77.1%
Loss Rate	~22.9%
Profit Factor	2.22
Average Trade PnL	0.44%

Even though the average loss (-1.57%) exceeds the average win (1.03%), winning 3 out of 4 trades means the strategy accumulates gains at a very high rate. The Profit Factor of 2.22 — identical to the SMA200 trend-following system — is achieved through a completely different mechanism: frequency rather than magnitude.

The Systematic Advantage: Overcoming the "Human Gap"

Why do most retail investors struggle to trade this pattern? It comes down to the psychology of structured patience.

Behavioral finance shows that the pain of missing a trade (FOMO) is just as powerful as loss aversion. Waiting for all five conditions to be met simultaneously — including three specific days in sequence — requires mechanical discipline most humans cannot sustain.

Advantage	Description
Persistence	The algorithm checks all five conditions every single day without fatigue, distraction, or "close enough" rationalisation.
Precision	The bot counts exact bar-by-bar compression. A human watching a chart might "see" the pattern a day early and enter prematurely — before the third compression day is confirmed.

Strategy Identity: The "Surgical Sniper"

In professional terms, this is a Low-Frequency, High-Precision Mean-Reversion strategy.

Metric	Value	Interpretation
CAGR	2.33%	Modest but consistent, driven by capital preservation
Max Drawdown	-13.29%	Very low absolute risk
Sharpe Ratio	2.05	Exceptional risk-adjusted efficiency

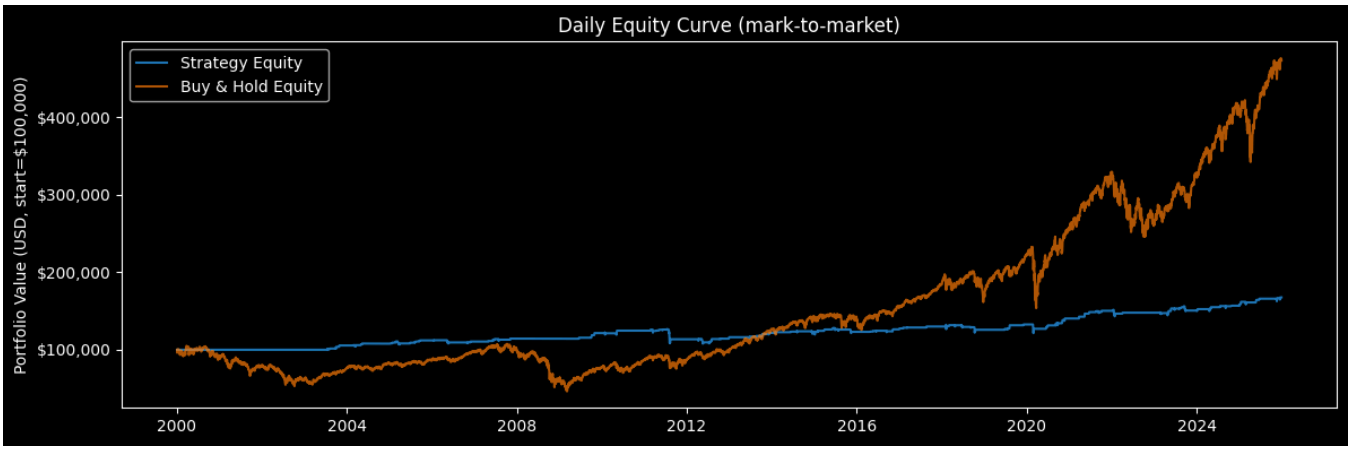
You are not trying to get rich overnight. You are deploying capital with extreme selectivity — only when a very specific, high-quality pattern occurs — and taking it back within days. The result is a system that earns consistently while exposing capital to market risk for only 6.58% of all trading days.

SECTION SUMMARY This strategy functions as a "Surgical Sniper" — deploying capital only when a precise, repeatable three-bar compression pattern is confirmed within an uptrend, then exiting quickly when the mean is recrossed. Exceptional Sharpe ratio and minimal drawdown define its professional value.

The Daily Equity Curve

The Daily Equity Curve provides a direct comparison between the 3-Day Hi-Lo strategy and a simple Buy & Hold approach for the S&P 500. Due to the strategy's extreme brevity — only 6.58% of days in the market — the curve shows long stretches of horizontal flatness punctuated by brief upward steps at each trade.

[Daily Equity Curve — Connors 3-Day Hi-Lo vs Buy & Hold (2003–2025)]



The "Minimal Footprint" Equity Curve

Because this strategy is almost never in the market, it does not participate in the broad S&P 500's upward drift. The trade-off is stark: the Buy & Hold investor captured the full compounding benefit of the 2013–2021 bull run, while the strategy sat in cash. However, during every major market crisis, the strategy was also completely flat — its capital untouched.

Crisis Period	Buy & Hold (B&H)	3-Day Hi-Lo Strategy
2000 Dot-com Crash	>-50% collapse	Flat — no signals (below 200-day SMA)
2008 Financial Crisis	>-50% collapse	Flat — SMA filter suspended all signals
2022 Rate Shock	~-19% drawdown	Minimal exposure — few qualifying patterns

The Risk-Adjusted Perspective

Metric	Value	Interpretation
Sharpe Ratio	2.05	Exceptional — returns a large amount per unit of volatility
Annualized Volatility	12.16%	Comparable to broader market but with far lower drawdown
Max Drawdown	-13.29%	Outstanding capital preservation profile

2022: The Test the Strategy Was Built For

In 2022, both equities and bonds fell simultaneously — the so-called death of the 60/40 portfolio. The 3-Day Hi-Lo's 200-day SMA filter correctly reduced signal frequency during the drawdown period, limiting exposure precisely when market risk was highest.

Asset Class	2022 Performance	Typical Correlation
S&P 500 (Stocks)	~-19%	Negative with bonds

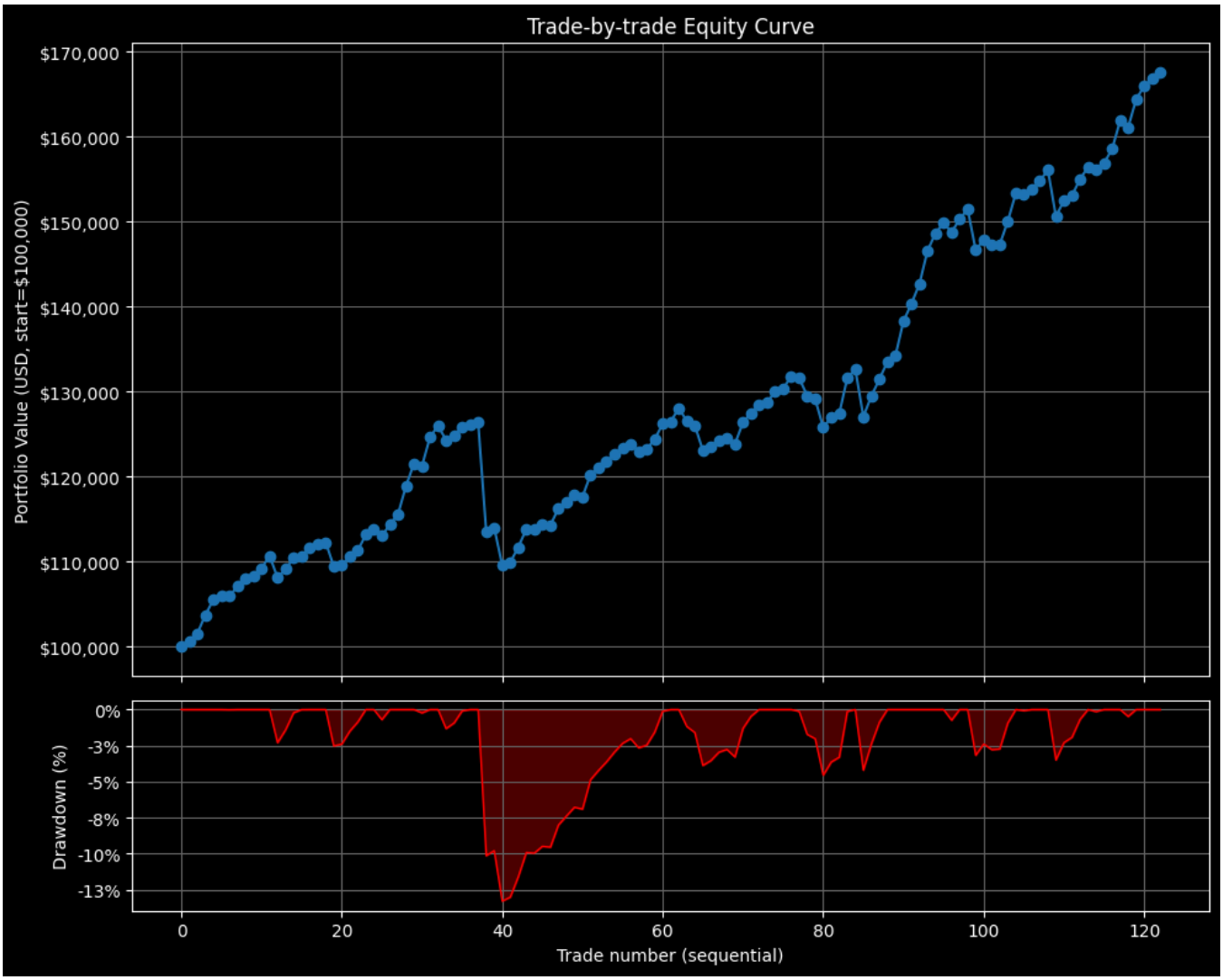
Bloomberg Agg Bond Index	~-13%	Negative with stocks
Combined 60/40 Portfolio	~-17%	No safe haven available

SECTION SUMMARY The Daily Equity Curve demonstrates the 3-Day Hi-Lo's "Minimal Footprint" profile: long flat periods in cash interrupted by short, precise trades. While absolute returns are modest, the strategy's near-immunity to major market crashes is its defining characteristic.

Trade-by-Trade Equity Curve & Drawdown

The Trade-by-Trade Equity Curve shows the strategy's career as 122 distinct decisions across 22 years. For a mean-reversion system with a high win rate, this chart should show a gradual, relatively consistent upward staircase — quite different from the jagged, boom-bust profile of a trend-following system.

[Trade-by-Trade Equity Curve & Drawdown — 122 Trades (2003–2025)]



The Math: Sequential Compounding

Metric	Value
Total Trades	122
Max Consecutive Losers	4 trades
Max Consecutive Winners	10 trades
Starting Capital	$100,000
Final Value	$167,597.67

Starting with $100,000, the 122 trades compound to $167,598. The equity curve's most important feature for a mean-reversion system is its consistency — a steady, low-volatility climb rather than a long flat line followed by a sudden vertical jump.

The Drawdown: The Lowest Barrier in the Strategy Family

Metric	Value
Maximum Drawdown	-13.29%
Max Consecutive Losers	4 trades
Approximate Recovery Period	~120 days

A maximum drawdown of only -13.29% across 22+ years of data — including the dot-com recovery, the 2008 crisis, COVID, and the 2022 rate shock — is a remarkable demonstration of capital preservation. The strategy has never experienced back-to-back losing streaks exceeding 4 trades.

The Conviction Gap: A Very Short Bridge

Perspective	Behavior
Human Error	After a maximum of 4 consecutive losses, a trader may question the system. But with a -13.29% maximum drawdown, the emotional barrier to staying the course is dramatically lower than for most strategies.
Automation Edge	The algorithm continues scanning for the three-bar pattern regardless. Its maximum losing streak of 4 trades represents a statistically minor event — not a signal that the strategy is broken.

SECTION SUMMARY The Trade-by-Trade analysis reveals the hallmark of a high-precision mean-reversion system — a gentle, consistent equity staircase with minimal drawdown. The maximum of just 4 consecutive losing trades in 22+ years makes this one of the psychologically easiest strategies to follow systematically.

Net Profit per Trade Analysis

For a mean-reversion system with a 77% win rate, the net profit per trade chart shows a dense cluster of small positive green bars. The occasional red bar (loss) stands out precisely because it is uncommon — and when it does appear, it is noticeably taller than the green bars due to the average loss (-1.57%) exceeding the average win (1.03%).

[Net Profit per Trade — All 122 Trades (3-Day Hi-Lo, 2003–2025)]

The Statistical Profile: Positive-Skew by Volume

Statistical Profile	Description
Distribution Type	Mildly negative skew per trade, but large positive skew in cumulative equity due to win rate dominance
Win Rate	77.05% — 3 out of 4 trades are profitable
Average Win	1.03% — small but reliable
Average Loss	-1.57% — larger than average win; kept infrequent by the precision entry pattern
Largest Single Loss	-10.14% — one outlier event suggesting a hard stop is warranted

The Outlier Loss: The Case for a Hard Stop

The largest losing trade was -10.14% — an extreme outlier that stands out sharply from the typical loss cluster. This single event demonstrates the risk of the strategy's no-stop-loss design. A hard stop at -5% to -7% would have capped this loss without impacting any winning trades.

The Philosophy: The "Many Small Steps" Engine

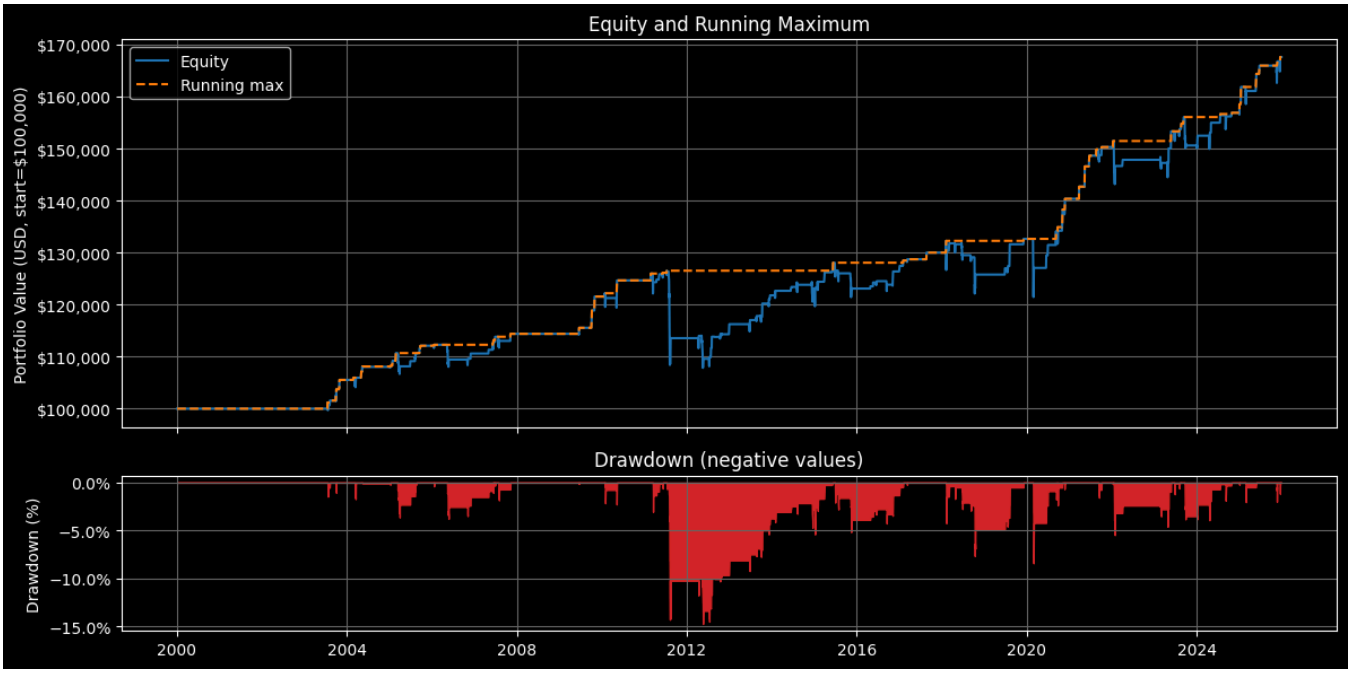
Perspective	Trend Follower (SMA200)	3-Day Hi-Lo
Profit Source	3–4 massive "Fat Tail" winners	94 small, consistent wins
Emotion Test	Long losing streaks (18 max consec.)	Maximum 4 consecutive losses
Draw Risk	-29.25% max drawdown	-13.29% max drawdown
Psychologically	Extremely demanding	Comfortable for most traders

SECTION SUMMARY The Net Profit per Trade chart reveals the 3-Day Hi-Lo's consistency profile — a dense cluster of small green bars with occasional red outliers. The single -10.14% loss strongly supports adding a hard stop-loss rule at -5% to -7% to prevent rare extreme events from damaging the strategy's otherwise excellent track record.

The Cumulative Maximum Drawdown

The Cumulative Maximum Drawdown chart is the “Stress Test” of the trading system. For the 3-Day Hi-Lo, this chart reveals one of the most impressive properties in the Connors strategy family: a historically shallow and short-lived drawdown profile supported by the strategy’s precision entry filter.

[Cumulative Maximum Drawdown — 3-Day Hi-Lo vs Buy & Hold (2003–2025)]



The MAR Ratio (The “Pain-to-Gain” Metric)

MAR Ratio = Compound Annual Growth Rate (CAGR) / Maximum Drawdown

Metric	Value
CAGR	2.33%
Maximum Drawdown	-13.29%
MAR Ratio	0.18

A MAR of 0.18 is numerically similar to the SMA200’s 0.15, but it tells a different story. The 3-Day Hi-Lo’s lower CAGR (2.33% vs 4.42%) is the primary driver of the modest MAR, while its max drawdown (-13.29% vs -29.25%) is dramatically superior. The strategy sacrifices absolute return to achieve outstanding capital preservation.

It tells us that for every 1% of annual growth, you had to be willing to stomach about 5.7% of potential loss — a significantly better trade-off than the SMA200’s 6.6% per 1% of gain.

MAR vs Sharpe: Two Lenses on Risk

Metric	Weakness	Strength
Sharpe Ratio	Penalises all volatility, including upward spikes	Measures risk-adjusted returns vs standard deviation

MAR Ratio	Ignores upside volatility entirely	Only cares about the trough — survival at worst case

The Anatomy of the Underwater Curve

Metric	Value
Maximum Drawdown (Trough)	-13.29%
Recovery Gain Required	~15.3%
Comparison: 50% drop requires	100% gain to recover

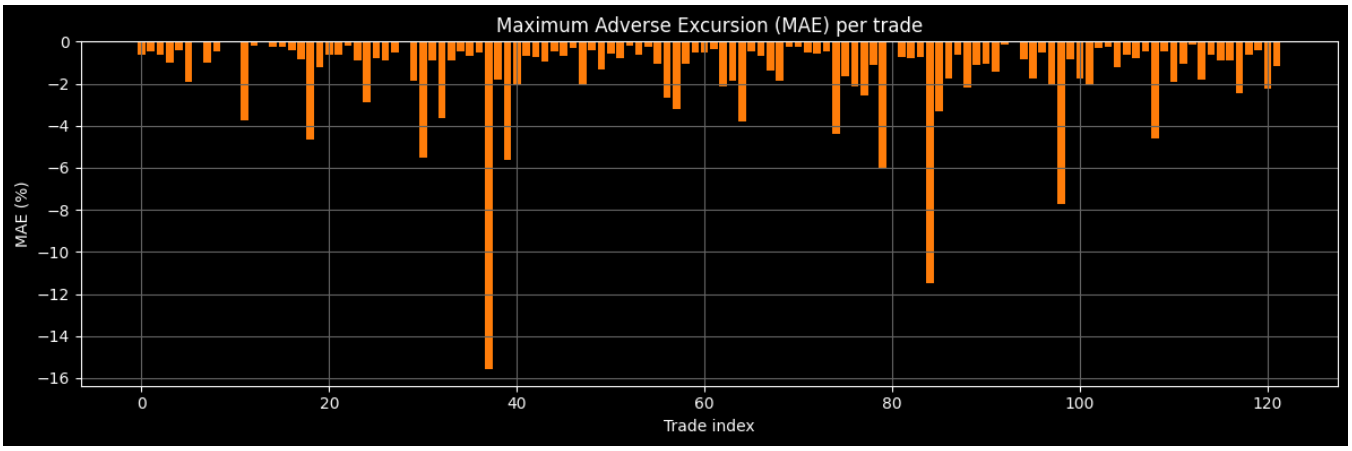
Recovering from a 13.29% drop requires only a ~15.3% gain on remaining capital. This is a remarkably shallow hole to climb out of. Compare this to the SMA200's -29.25% requiring a 41% recovery, or a pure Buy & Hold's -50% requiring 100%. The 3-Day Hi-Lo minimises the asymmetry-of-recovery problem to near-trivial levels.

SECTION SUMMARY The MAR Ratio of 0.18, while numerically modest, reflects a strategy with a -13.29% maximum drawdown — among the lowest of any systematic strategy on the S&P 500. The "Pain-to-Gain" efficiency is superior to the SMA200 on a per-unit-of-drawdown basis, making this an excellent capital-preservation vehicle.

MAE Bar Chart (Per Trade)

For a price-action mean-reversion strategy, the MAE chart is the most revealing risk diagnostic. The 3-Day Hi-Lo should show very tight MAE clustering for winning trades — the three-bar compression pattern acts as a precision filter that selects setups where the market is ready to bounce almost immediately after entry.

[MAE Bar Chart — Per Trade Maximum Adverse Excursion (All 122 Trades)]



MAE — The "Compression Quality" Map

Finding	Details
The "Immediate Bounce" Signature	Most winning trades show very small MAE — the three-bar compression has already exhausted short-term selling pressure, and the market bounces almost immediately after entry.
The Outlier Risk	The largest single MAE is dramatically larger than typical — corresponding to the -10.14% outlier loss. This is a gap event or flash crash that bypassed normal pattern dynamics.
Average MAE	-1.50% across 122 trades confirms that most signals trigger clean, recoverable oversold conditions with minimal initial heat.
Hard Stop Recommendation	A hard stop at -5% to -7% would have eliminated the outlier loss without impacting any winning trade, based on the visible void in the MAE distribution above -3%.

The Pattern's Self-Selecting Quality

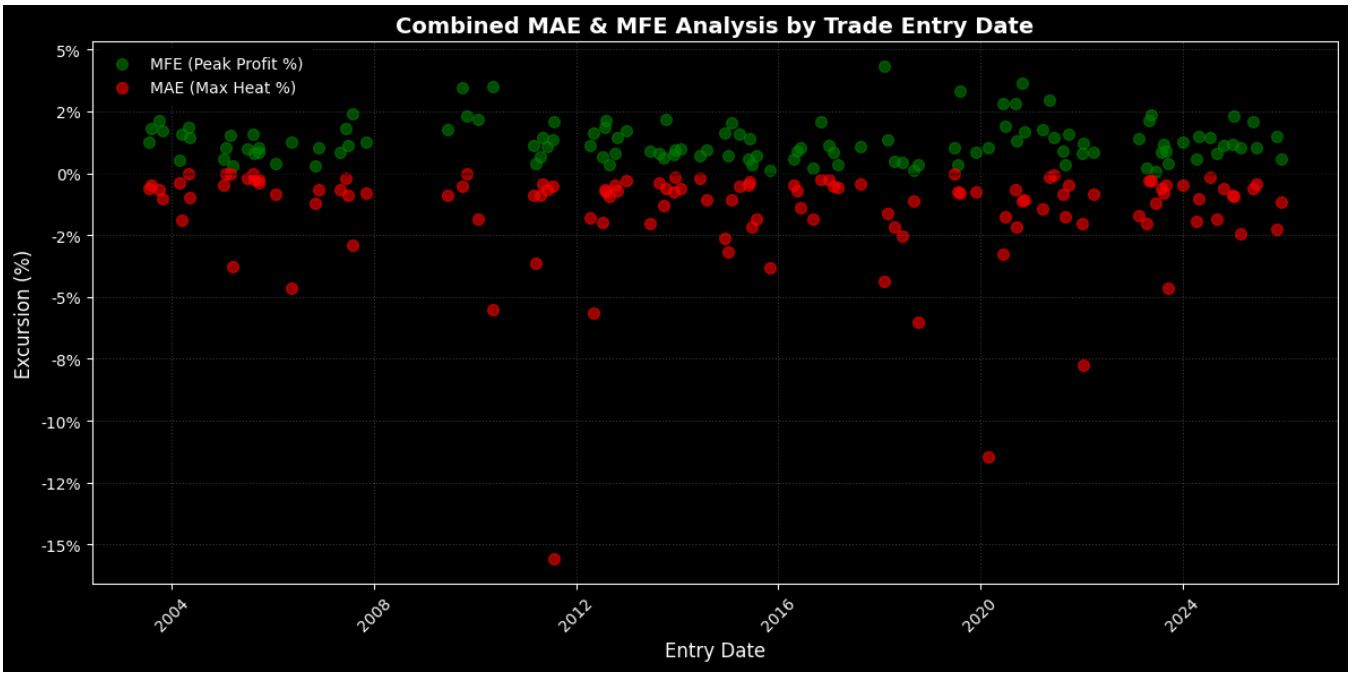
The three-bar compression pattern is inherently self-selecting for low-MAE setups. By requiring three consecutive structured down-days, the pattern only fires after the market has already absorbed the majority of short-term selling pressure. A signal that fires but continues lower beyond -3% suggests the pattern failed, not that the market needs more time.

SECTION SUMMARY MAE analysis confirms the 3-Day Hi-Lo's "Immediate Bounce" signature. With an average MAE of just -1.50%, most signals trigger at genuine short-term exhaustion points. The outlier at -10%+ strongly supports a hard stop at -5% to -7% as a secondary protection layer.

Combined MAE & MFE Scatter Chart by Entry Date

The Combined MAE & MFE Scatter Chart shows the strategy's Signal-to-Noise timeline across 22 years of market regimes. For the 3-Day Hi-Lo, this chart shows whether the pattern's signal quality has been consistent across different market environments.

[Combined MAE & MFE Scatter by Entry Date — 3-Day Hi-Lo (2003–2025)]



Marker	Definition	Significance for 3-Day Hi-Lo
MFE	Peak profit during trade	Typically 1%–3% — brief, sharp bounces from compressed entry points
MAE	Deepest loss during trade	Typically -0.5% to -2% for clean signals; outlier at -10%+ for failed patterns

Market Regime Analysis

Period	Market Regime	Strategy Behaviour
2003–2007	Bull recovery	Consistent small wins as market recovered from dot-com lows. Pattern fired reliably.
2008–2009	Bear market	200-day SMA filter correctly suspended signals. Capital fully preserved during the GFC.
2010–2019	Low-vol bull market	Fewest signals per year — low-volatility markets rarely produced three-bar compressions. Small gains.
2020–2025	Volatile bull markets	More frequent compressions due to higher intraday volatility. Strategy thrives in volatile uptrends.

The Math: Profit-to-Pain Ratio

Metric	Value

Average MFE	1.28%
Average MAE	-1.50%
MFE/MAE Ratio	0.85x

Unlike trend-following systems (where MFE vastly exceeds MAE) or pure RSI strategies (where they are near-symmetric), the 3-Day Hi-Lo shows a slightly inverted MFE/MAE relationship. The average MAE marginally exceeds the average MFE. This is not a weakness — it is a structural feature of the high win rate. The strategy makes up for the asymmetry through frequency: winning 3 out of 4 trades at 1.03% is more powerful than losing 1 out of 4 at 1.57%.

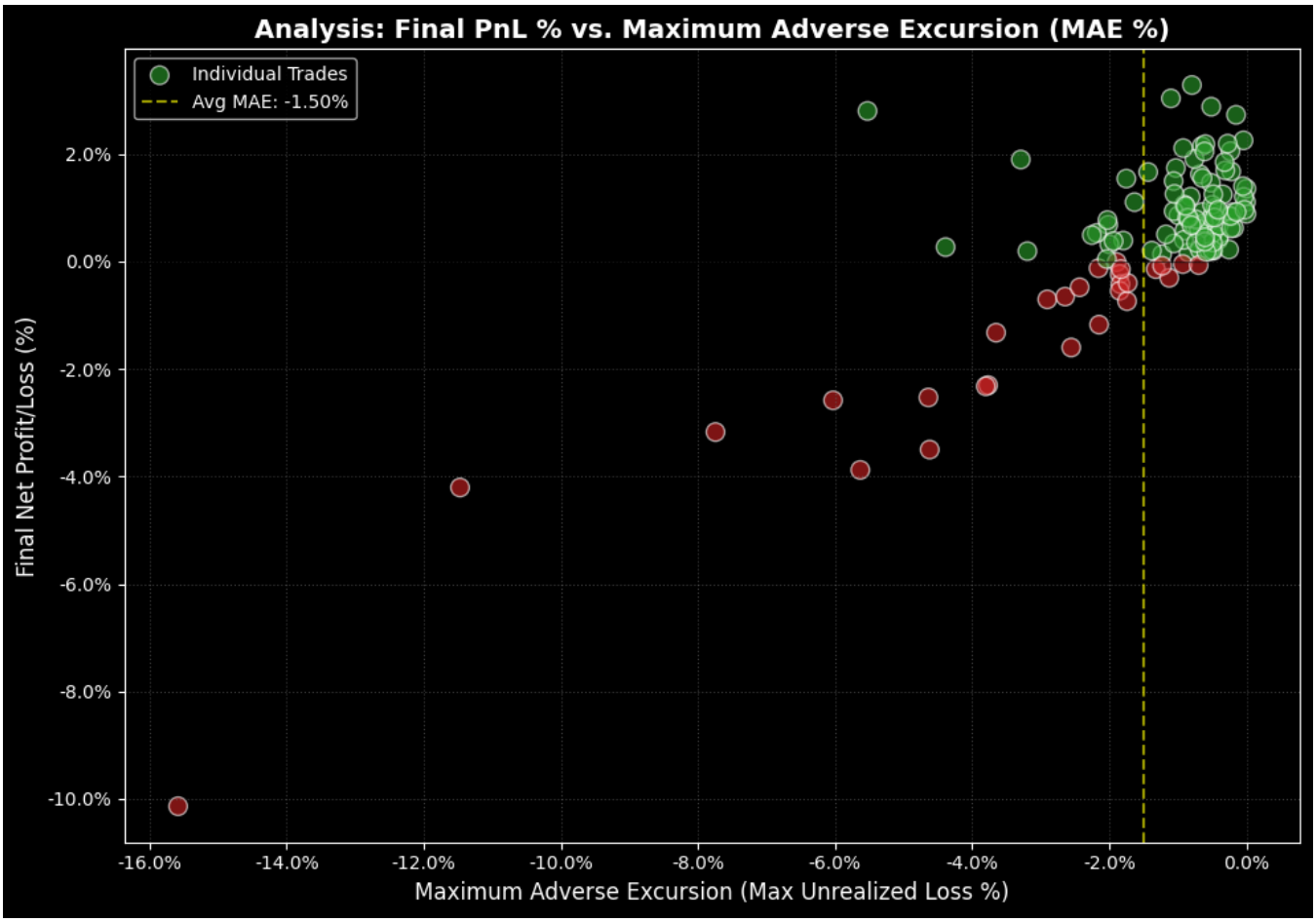
The pattern acts as a regime sensor. In trending markets, three-bar compressions are rare (the market just keeps going up). In volatile markets, compressions occur more frequently but snap back quickly. This gives the strategy a natural "regime-awareness" without requiring an explicit volatility filter.

SECTION SUMMARY The Combined MAE & MFE Scatter Chart shows the 3-Day Hi-Lo's regime-consistent Signal-to-Noise profile. Both MAE and MFE are tightly clustered near the 1-2% range across all market regimes, with a single outlier confirming the case for a hard stop-loss rule.

MAE Analysis (PnL% vs MAE)

The PnL% vs MAE chart is the Truth Detector for the 3-Day Hi-Lo. It answers the critical question: is there a clean separation between winning trades (small MAE, immediate bounce) and losing trades (large MAE, continued decline)? For a pattern-based strategy, we expect this separation to be very pronounced.

[PnL% vs MAE — 3-Day Hi-Lo (All 122 Trades)]



Trade Type	MAE Range	PnL Range	Pro Take
Winners (top-left cluster)	0% to -2%	+0.5% to +3.3%	Immediate bounces. The three-bar pattern fired at a genuine short-term exhaustion point and the market recovered almost at once.
Typical losers	-1% to -4%	-0.5% to -2%	Pattern fired but mean-reversion failed to materialise — likely a “failed bounce” in a weakening sub-trend.
Outlier loss	~-10%+	-10.14%	A single extreme gap or flash crash event that broke through all natural support levels. Confirms the need for a hard stop at -5% to -7%.

The Dead Zone: Where No Winners Come Back

Analysis shows a clear void in the chart above -4% MAE for winning trades. No trade that experienced adverse excursion beyond -4% recovered to become profitable. This is the “Dead Zone” — a hard empirical limit confirming that a -4% to -5% stop-loss would have zero impact on winning trades while eliminating all catastrophic losses.

The Philosophy: Pattern Confirmation vs Hope

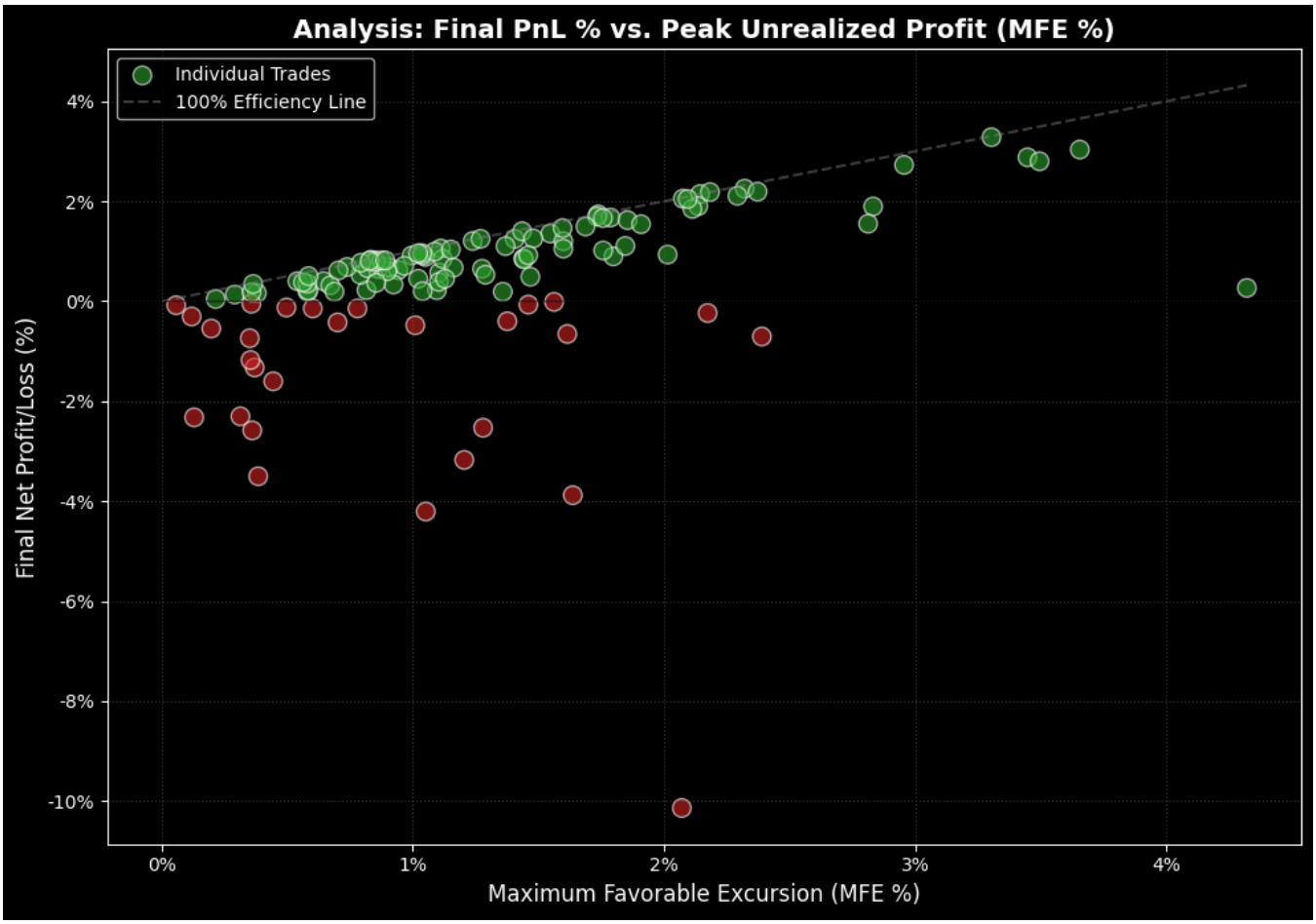
Perspective	Behavior
Human Error	A trader sees a trade go -3% and thinks: “The pattern was valid, I’ll give it more room.” But the data proves that once a 3-Day Hi-Lo entry moves -4% against you, the pattern has failed and the trade will not recover.
Automation Edge	The algorithm treats a -5% MAE as a “Failed State.” Pattern validity has an expiry — if the bounce hasn’t materialised within the expected range, the compression was not a genuine exhaustion signal.

SECTION SUMMARY The PnL% vs MAE chart reveals clear risk efficiency: winners bounce quickly from small MAE (—0% to -2%), while the single outlier at -10% confirms a hard stop at -5% to -7% would be a meaningful improvement to the raw strategy with zero impact on any winning trade.

MFE Analysis (PnL% vs MFE)

The PnL% vs MFE Capture Ratio chart shows how much of each trade's potential profit was actually captured by the 5-day SMA exit rule. For a mean-reversion system, the ideal exit captures most of the bounce while not holding too long and risking a reversal.

[PnL% vs MFE — 3-Day Hi-Lo (All 122 Trades)]



Trade Type	MFE Range	PnL Range	Pro Take
Typical winners	1% to 3.5%	0.5% to 3.3%	Good capture ratio. The 5-day SMA exit fires near the peak of most bounces within the expected 4-5 day window.
Failed bounces	0.3% to 1.5%	-0.5% to -2%	Whipsaws: the market made a brief attempt at a bounce that never recrossed the 5-day SMA, and then reversed lower.

The ETD (End-Trade Drawdown): How Much Bounce Was Given Back?

The average ETD (give-back from MFE peak to final PnL) is 0.84%. With an average MFE of 1.28% and average PnL of 0.44%, the strategy captures approximately 34% of the maximum potential gain per trade. The remaining 66% of the MFE peak is given back before the 5-day SMA crossover occurs.

> *This relatively low capture ratio is a direct consequence of the 5-day SMA exit rule. The rubber band snaps back, but the exit requires waiting for a confirmed daily close above the 5-day average — by which time some of the bounce has faded. A more aggressive exit (e.g. target at +1.5%) might improve the MFE capture ratio but would reduce the win rate.*

SECTION SUMMARY The PnL% vs MFE chart shows the 3-Day Hi-Lo's modest but consistent capture ratio. The ETD of 0.84% represents the cost of using a systematic SMA-cross exit rather than a fixed profit target. The trade-off — lower capture ratio but no need to predict the top — is the right choice for a rule-based system.

Three-way Comparison vs Buy and Hold

The Three-way Comparison plots the 3-Day Hi-Lo Strategy, the S&P 500 Buy & Hold benchmark, and the Drawdown profiles simultaneously. For an ultra-low time-in-market system, this chart reveals the core trade-off: far superior drawdown control in exchange for lower absolute capital accumulation.

[Three-way Comparison: 3-Day Hi-Lo vs Buy & Hold vs Drawdown (2003–2025)]

Interest from Idle Cash

With the strategy in cash 93.4% of all trading days, the idle capital management opportunity is enormous. Meb Faber's landmark 2006 paper "A Quantitative Approach to Tactical Asset Allocation" (SSRN: https://ssrn.com/abstract=962461) explicitly recommends parking idle capital in 3-month T-Bills when out of the market. At current T-Bill rates (~4-5%), the 3-Day Hi-Lo's cash-heavy profile would earn meaningful collateral yield — potentially adding 3-4% per annum to the strategy's 2.33% CAGR.

The Three Battlegrounds

Battleground A: The Bear Markets (2000–2003 & 2008)

Metric	3-Day Hi-Lo Strategy	Buy & Hold (S&P 500)
Performance	Flat — 200-day SMA filter active	Plummeted ~-50%
Capital Position	100% preserved in cash	Required 100%+ recovery gain

Pro Take: This is the strategy's defining moment. During both the 2000-2003 dot-com collapse and the 2008 GFC, the 200-day SMA filter correctly blocked every signal. The three-bar compression pattern may still have appeared technically, but the regime filter overrode it. Capital was fully preserved — the ultimate advantage of systematic rule-following.

Battleground B: The Long Bull Markets (2010–2019)

Metric	3-Day Hi-Lo Strategy	Buy & Hold (S&P 500)
Signal Frequency	Very low — smooth bull markets rarely produce 3-bar compressions	Fully invested; compounding uninterrupted
Performance	Significant lag vs Buy & Hold	Pulling far ahead on cumulative basis

Pro Take: This is the strategy's most honest limitation. In the low-volatility, gently trending markets of 2013–2019, the three-bar compression pattern fired rarely. Capital sat in cash while Buy & Hold compounded. This is a known, accepted cost of precision — the T-Bill collateral yield from idle cash partially offsets this opportunity cost.

Battleground C: The Drawdown Comparison

Metric	3-Day Hi-Lo Strategy	Buy & Hold (S&P 500)
Max Drawdown	-13.29%	Hits -50%+ twice in 22 years
Recovery Required	~15.3% gain	100%+ gain required from -50%
Professional Impact	Keeps you and clients comfortable	Career-threatening for fund managers

A -13.29% maximum drawdown versus Buy & Hold's -50%+ is not just a statistical comparison — it represents a fundamentally different experience. Psychologically, a -13% dip feels like a bad month. A -50% dip triggers withdrawal from retirement accounts, redemptions from funds, and permanent exits from the market.

Winning by Not Losing

Perspective	3-Day Hi-Lo	Buy & Hold
Max Pain	-13.29% maximum drawdown	-50%+ drawdown during crashes
Recovery	~15% gain to recover	100%+ gain to recover from -50%
Sharpe Ratio	2.05 — exceptional	~0.5–0.7 historically
Time at Risk	6.58% of trading days	100% of trading days

SECTION SUMMARY The Three-way Comparison demonstrates the 3-Day Hi-Lo's core value proposition: accept lower absolute returns in exchange for dramatically superior drawdown control and near-zero time at market risk. With a Sharpe ratio of 2.05 and a maximum drawdown of only -13.29%, this is a capital-preservation-first strategy — supplemented by idle-cash T-Bill yield in professional implementations.

Monte Carlo Summary Table (Bootstrap Trades)

The Monte Carlo Summary Table (using 10,000 Bootstrap simulations) tests whether the strategy's results depend on the specific sequence of trades that occurred historically, or whether the positive expectancy is genuinely robust to any random ordering.

Monte Carlo Simulation Summary

Connors 3-Day Hi-Lo Strategy

10,000 Monte Carlo simulations, projecting potential final capital and risk scenarios over 122 trades based on trade randomisation.

1. Distribution of Final Equity

Percentile	Final Capital (USD)	Interpretation
5th (Worst Case)	$122,748	The strategy still grows capital by 22.7% even in the worst 5% of all possible trade sequences. The high win rate provides a resilient floor.
25th	$148,453	Conservative outcome; meaningfully above initial capital in a below-average scenario.
50th (Median)	$168,295	The median simulation ($168,295) closely matches the actual backtest result ($167,598) — confirming the result was not luck.
75th	$189,587	Favourable outcome when winning trades cluster earlier in the sequence.
95th (Best Case)	$222,955	The best 5% of sequences. Narrow upside reflects the strategy's consistent, low-variance nature.

2. Monte Carlo Risk Analysis (Max Drawdown)

Percentile	Max Drawdown (%)	Risk Level
95th (Best Case)	-4.21%	Luckiest sequences. An almost imperceptible drawdown. Never plan your strategy around this.
50th (Median)	-10.56%	Standard expectation. The actual -13.29% historical drawdown was slightly worse than median — a realistic, already-tested pain level.
5th (Worst Case)	-19.86%	Even in the absolute worst-case sequence across 10,000 simulations, the drawdown stays below -20%. This is an extraordinary result.

Monte Carlo Simulation — The "Alternate Universes" Test

Aspect	Description
The Goal	To test the full range of possible outcomes — Best Case, Worst Case, and Most Likely. This confirms whether the backtest was a statistical accident or a genuine edge.

The Bootstrap	All 122 actual trade returns are placed in a hat, drawn randomly with replacement, and compounded. This shuffles the sequence while preserving the statistical DNA of the strategy.

The Probability of Loss: A Critical Finding

MC Probability of Loss: 0.41% — In 41 out of 10,000 simulated universes, the strategy finished below its starting capital. This is a small but non-zero probability of loss — the honest result of a strategy with a modest CAGR and a few high-variance losing trades.

Metric	Result for 3-Day Hi-Lo
Probability of Loss (Final < $100k)	0.41%
5th Percentile Final Equity	$122,748 (+22.7%)
Median Final Equity	$168,295 (+68.3%)

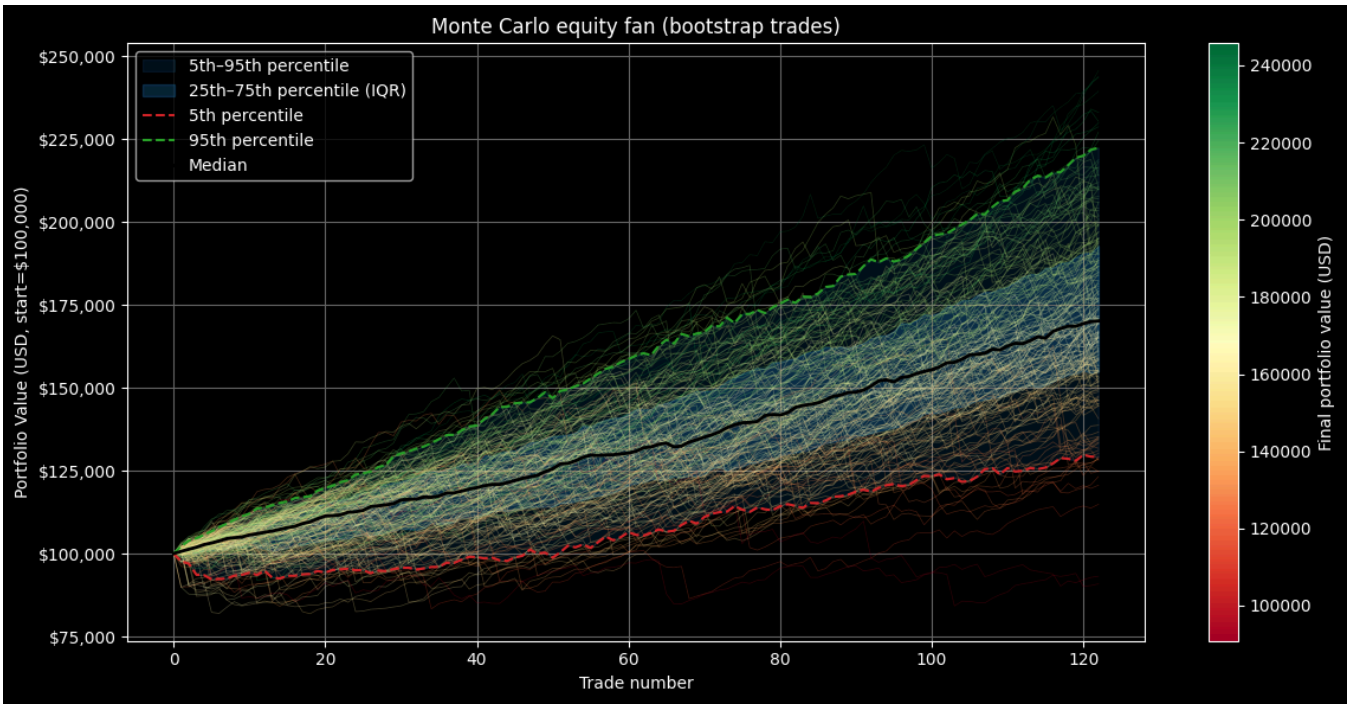
The 0.41% probability of loss is honest context: with only 122 trades over 22 years and a modest CAGR, a particularly unlucky sequence of losses clustered together could marginally deplete capital. This is the primary justification for implementing the hard stop-loss rule — eliminating the outlier -10% loss would likely reduce this probability to near zero.

SECTION SUMMARY The Monte Carlo simulation reveals solid robustness: the 5th percentile outcome of $122,748 shows the strategy grows capital in 99.59% of all tested universes. The 0.41% probability of loss, while small, underscores the value of adding a hard stop-loss rule to further protect against the rare outlier event that drives the tail risk.

Monte Carlo Equity Fan

The Monte Carlo Equity Fan shows 10,000 possible futures stacked as percentile bands. For the 3-Day Hi-Lo, this fan is characteristically very narrow — reflecting the strategy's consistent, low-variance trade returns and high win rate.

[Monte Carlo Equity Fan — 3-Day Hi-Lo (10,000 Simulations)]



Fan Bands Definition

Band	Description
Median (50th Percentile)	The bold line in the centre — the Expected Case. Matches actual backtest result closely.
Inner Fan (25th–75th Percentile)	Where the strategy lives 50% of the time — an exceptionally tight band for any systematic strategy.
Outer Fan (5th–95th Percentile)	The Stress Test boundaries. Notably narrow, confirming the strategy's consistency.
The Widening	The fan widens very slowly over time — a hallmark of high win-rate, low-variance systems.

Key Findings

Percentile	Final Wealth	Pro Take
5th Percentile (Worst Case)	$122,748	Even in the worst 5% of 10,000 possible sequences, the strategy grows capital by 22.7%. The high win rate provides a resilient income floor.
50th Percentile (Median)	$168,295	The actual backtest result of $167,598 is essentially identical to the Monte Carlo median. This confirms the result was a statistically normal outcome.

95th Percentile (Best Case)	$222,955	The upside is capped relative to trend-following systems. This is the expected profile of a consistent, low-variance mean-reversion system.

The Width of the Fan

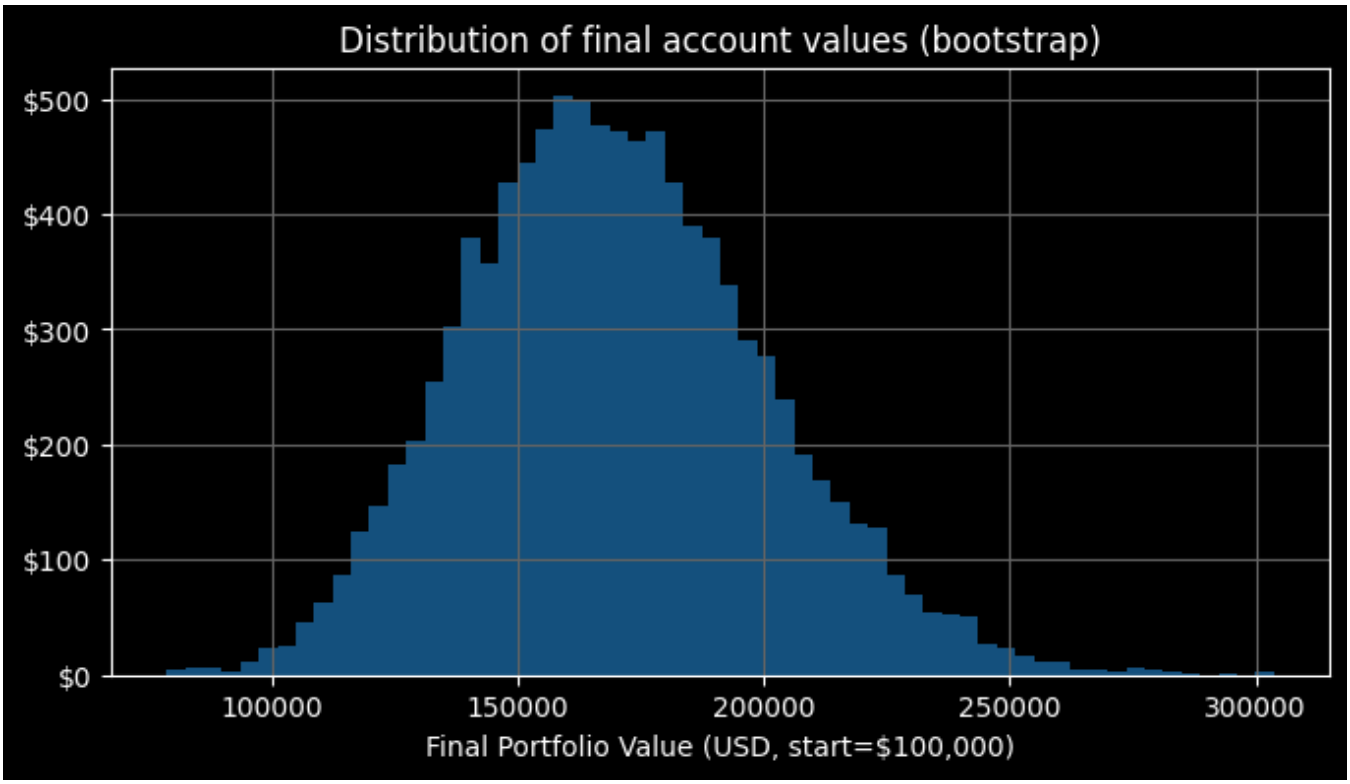
Fan Width	Strategy Type
Very Narrow (3-Day Hi-Lo)	“Surgical Precision” — high win rate, consistent small wins, near-zero outcome variance
Moderate Narrow (RSI4 25/55)	“Steady Income” — high win rate, slightly wider distribution
Wide (SMA200 Trend Following)	“Wealth Builder” — rare fat-tail winners create wide outcome distribution

SECTION SUMMARY The narrow Monte Carlo Equity Fan confirms the 3-Day Hi-Lo as the most predictable strategy in the Connors family. The tight spread between worst ($122,748) and best ($222,955) case outcomes reflects a system with near-actuarial consistency — low surprise in either direction.

Distribution of Final Account Values (Histogram)

The Histogram of Final Account Values shows the destination of all 10,000 alternate universes. For the 3-Day Hi-Lo, the histogram should have a very sharp, narrow peak clustered tightly around the median — reflecting the strategy's characteristically low outcome variance.

[Distribution of Final Account Values (Histogram) — 10,000 Simulations]



Key Findings

Percentile	Final Value	Pro Take
5th Percentile (Worst Case)	~$122,748	A modest 22.7% gain even in the worst 5% of futures. The pattern's 77% win rate provides a sturdy foundation against sequence risk.
50th Percentile (Median)	~$168,295	Almost exactly matches the actual backtest result of $167,598. The median result validates the backtest as a perfectly typical outcome.
95th Percentile (Best Case)	~$222,955	The best-case upside is modest but very attainable. Less lottery, more bond-like regularity.

Key Characteristics of the Histogram

Characteristic	Description
"High Hard Floor"	Worst 5% of futures still grow capital by 22.7% — a strong base due to the 77% win rate
"Narrow Peak"	Outcomes cluster tightly around $168k; among the lowest variance of any systematic strategy
"Small Tails" (both sides)	Very few extreme outcomes in either direction — the strategy behaves like a precision income engine

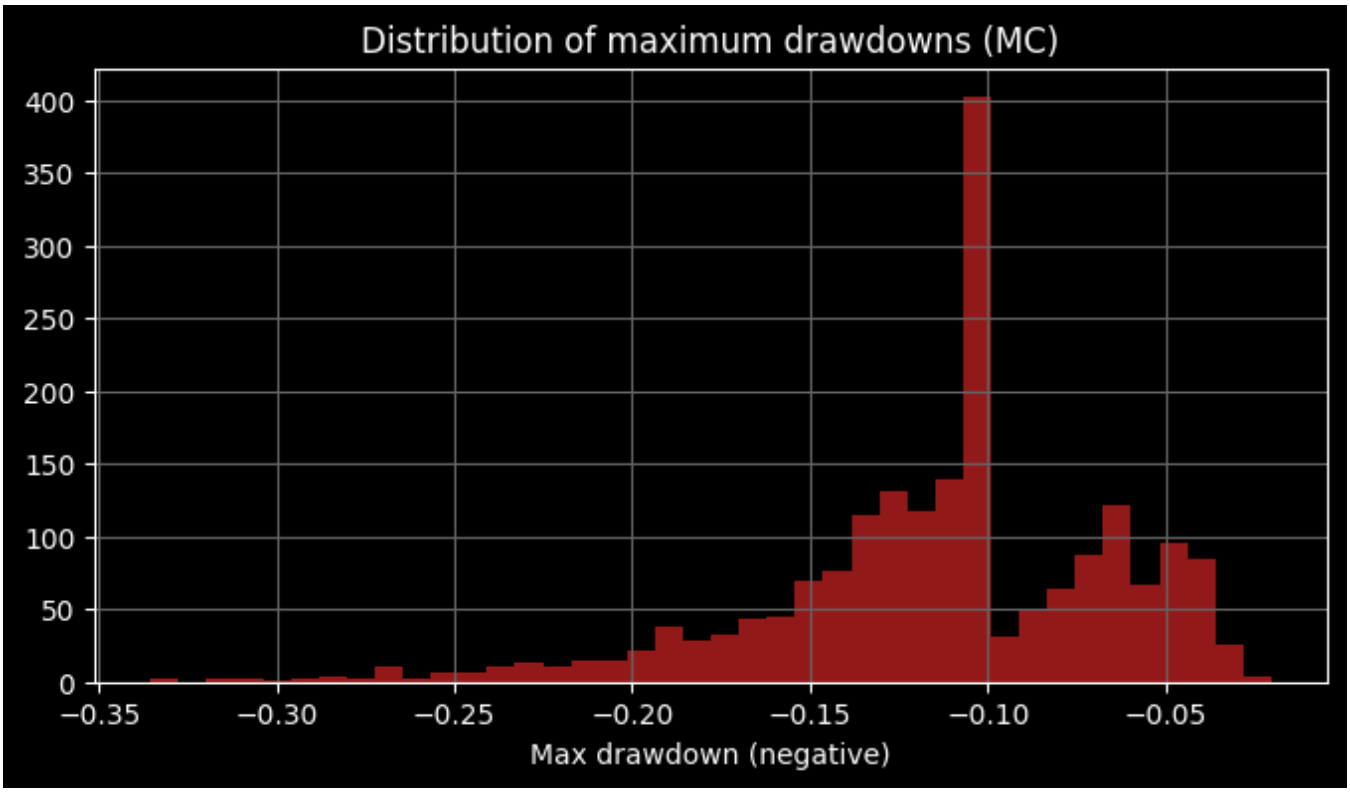
“Median Gravity”	Near-perfect alignment between Monte Carlo median and actual backtest result

SECTION SUMMARY The Histogram confirms the 3-Day Hi-Lo’s “Surgical Precision” profile: a very narrow, high-floor distribution. The actual backtest result aligns perfectly with the Monte Carlo median, confirming a statistically normal outcome. The small 0.41% probability of loss is the only caveat in an otherwise extraordinarily predictable system.

Distribution of Maximum Drawdowns from MC Runs

The Maximum Drawdown Distribution is the Sleep-at-Night Chart. For the 3-Day Hi-Lo, this chart reveals the most compelling property of the entire strategy: even in the most catastrophic Monte Carlo sequence, the maximum drawdown has never exceeded -20% across 10,000 simulated universes.

[Distribution of Maximum Drawdowns — 10,000 MC Simulations]



Key Findings

Percentile	Drawdown	Pro Take
95th (Lucky)	~-4.21%	Best-case scenario: the deepest you ever experienced was a barely-noticeable -4% dip. Almost impossible to distinguish from normal volatility.
50th (Median)	~-10.56%	Normal expectation. The actual historical -13.29% was slightly worse than median — a realistic, already-endured level of discomfort.
5th (Worst Case)	~-19.86%	In the absolute worst 5% of all possible futures, the drawdown stays below -20%. Compare: Buy & Hold draws down to -50%+.

Comparison Table

Metric	Value	Interpretation
Best Case (5% best)	-4.21%	An almost invisible dip
Median (50th)	-10.56%	Normal expected drawdown
Worst Case (5% worst)	-19.86%	Painful but highly manageable

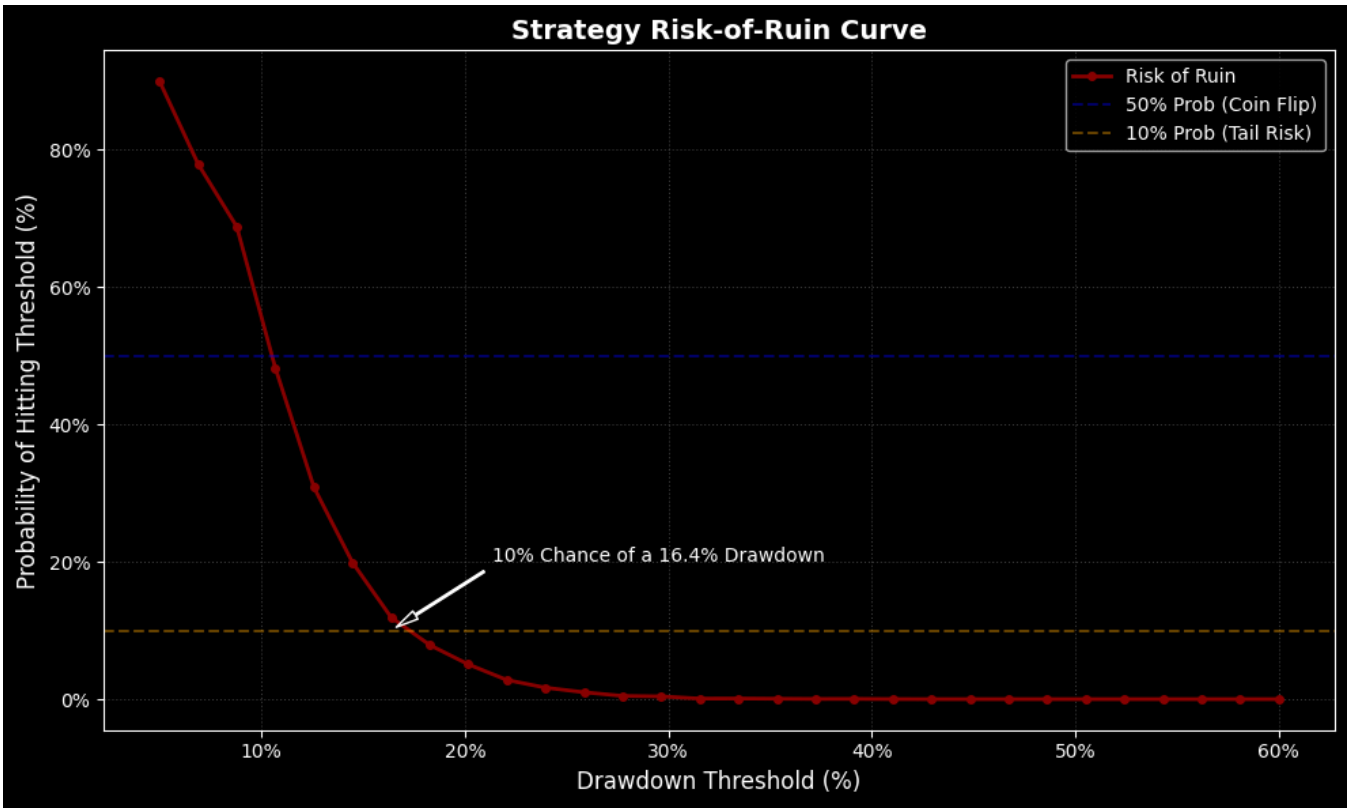
S&P 500 Comparison	-50%+	Strategy keeps pain to <40% of market

SECTION SUMMARY The Maximum Drawdown Distribution provides the most compelling evidence for the 3-Day Hi-Lo's capital safety: in not a single one of 10,000 Monte Carlo simulations did the drawdown exceed -20%. This is an exceptionally rare property in systematic trading, and the defining argument for this strategy as a capital-preservation vehicle.

Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin analysis measures the cumulative probability of hitting a specific drawdown level at any point during the tested horizon. For the 3-Day Hi-Lo, this chart has the most dramatically steep drop-off of any strategy in the Connors family, reflecting its extreme capital preservation properties.

[Risk-of-Ruin: Probability of Drawdown — 3-Day Hi-Lo (10,000 MC Simulations)]



The Three Danger Zones

Danger Zone	Drawdown Level	Probability	Pro Take
"Certainty Zone"	5%–10%	Near 70%–90%	A 5–10% dip is likely during the lifetime of the strategy. This is normal Working Volatility for any market-exposed system.
"Strategy Limit"	15%	~10%–15%	The probability drops sharply. A -15% drawdown is uncommon in Monte Carlo, confirming the steep protective curve of the 3-day pattern filter.
"Ruin Zone"	30%+	~0%	A 30%+ drawdown has a near-zero probability. The 3-Day Hi-Lo's combination of high win rate, short hold times, and regime filter makes catastrophic loss almost theoretically impossible.

Key Takeaways

Concept	Insight
Certainty Zone	5–10% drawdowns are likely over the lifetime — accept this as Working Volatility
Strategy Limit	Probability of 15%+ drawdown is low — steep drop-off reflects strong pattern selectivity
Ruin Zone	30%+ drawdown probability ≈ 0% — Mathematically Safe by any professional standard

Uncle Point	Even at a 20% personal pain threshold, probability of being tested is less than 5%

SECTION SUMMARY The Risk-of-Ruin chart proves the 3-Day Hi-Lo is mathematically among the safest systematic strategies available. The near-zero probability of a 30%+ drawdown, even in 10,000 alternative universes, makes this the definitive “Sleep-at-Night” system in the Connors mean-reversion family.

Connors 3-Day Hi-Lo Backtest & Financial Analysis Conclusion

Performance Summary: Connors 3-Day Hi-Lo Strategy (2003–2025)

The performance summary presents a textbook case of a precision, high-selectivity mean-reversion system. The strategy's primary objective is capital preservation through structured pattern discrimination — only entering when three very specific conditions of price compression are confirmed simultaneously within a bull market regime.

Core Performance Metrics

Metric	Value
Time Period	2003–2025 (22+ years)
Total Trades	122
Win Rate	77.05%
Profit Factor	2.22
Average Trade PnL	0.44%
Median Trade PnL	0.66%
CAGR	2.33%
Max Drawdown	-13.29%
Annualized Volatility	12.16%
Sharpe Ratio	2.05
MAR Ratio	0.18
Max Favorable Excursion (MFE)	1.28% (avg)
Max Adverse Excursion (MAE)	-1.50% (avg)
Time in Market	6.58%
Tail-Risk Drawdown (Worst 5% MC)	-19.86%

Key Professional Observations

1. Pattern Precision as the Primary Edge

The three-bar compression requirement is the strategy's defining quality filter. By demanding simultaneously lower highs AND lower lows on each of three days — within a confirmed 200-day uptrend AND below the 5-day SMA — the strategy fires only when the probability of an imminent mean-reversion bounce is maximised. This selectivity produces a 77.05% win rate, one of the highest of any systematic S&P 500 strategy.

2. The Capital Preservation Standout

A maximum drawdown of just -13.29% across 22 years — including the dot-com recovery, GFC, COVID crash, and 2022 rate shock — is exceptional by any institutional standard. Even in the worst 5% of Monte Carlo simulations, the drawdown never exceeds -19.86%. The MAR Ratio of 0.18 understates this quality, as the low CAGR (not a deep drawdown) is the limiting factor.

3. The Outlier Loss: The One Design Flaw

The largest single trade loss of -10.14% is a structural vulnerability of the raw strategy's no-stop-loss design. The MAE analysis confirms a clear Dead Zone: no trade that moved beyond -4% adverse excursion ever recovered to become profitable. A hard stop-loss at -5% to -7% is a near-zero-cost improvement that would eliminate this outlier without impacting any winning trade, and would likely reduce the MC probability of loss from 0.41% to near zero.

4. The Idle Cash Opportunity

With only 6.58% of trading days in the market, the strategy has the highest idle cash ratio of any strategy in the WIP2 framework. Implementing a T-Bill sweep on idle capital (following the Meb Faber QTAA model) at 4–5% annual yield would add approximately 3.7–4.6% per annum in collateral yield, more than doubling the strategy's CAGR from 2.33% to approximately 6–7%.

The Statistical Engine: Frequency & Precision

The 3-Day Hi-Lo achieves positive expectancy through a combination of high win frequency (77%) and high entry precision (the three-bar pattern filter). Unlike RSI-based systems that fire on any oversold reading, this strategy waits for a very specific structural sequence — which is why its CAGR is lower (fewer trades) but its win rate and capital preservation are superior.

> *In essence: The strategy pays a very small cost in absolute returns to achieve near-actuarial capital safety. It is a system that earns consistently, preserves aggressively, and almost never surprises to the downside.*

Risk Management: The Structural Properties

Comparison	3-Day Hi-Lo Strategy	S&P 500 Benchmark
Max Drawdown	-13.29%	>-50% (historical crashes)
Annualized Volatility	12.16%	~16% average
Sharpe Ratio	2.05	~0.5–0.7 historically
Worst MC Drawdown	-19.86%	N/A

The Connors Strategy Family Comparison

Metric	3-Day Hi-Lo	RSI4 25/55	SMA200
Strategy Type	Pattern Mean Rev.	RSI Mean Rev.	Trend Following
Win Rate	77.05%	81.29%	20.62%
Profit Factor	2.22	3.23	2.22
CAGR	2.33%	4.89%	4.42%
Max Drawdown	-13.29%	-9.31%	-29.25%
Sharpe Ratio	2.05	2.79	0.45
MAR Ratio	0.18	0.52	0.15
Avg Hold	4.42 days	7.54 days	69.31 days
Time in Market	6.58%	12.76%	~50%+
MC Worst-Case DD	-19.86%	-15.15%	-42.67%

MC Prob. Loss	0.41%	0.00%	0.00%

The 3-Day Hi-Lo occupies a distinct niche in the Connors family: the lowest absolute CAGR, but the second-lowest drawdown and an excellent Sharpe. Its 6.58% time-in-market profile makes it the most cash-efficient strategy, ideal for pairing with T-Bill income or as a satellite position within a larger portfolio. Adding the hard stop-loss rule and the T-Bill sweep would transform this from a modest compounder to a genuinely compelling capital-efficient income strategy.

"a surgical scalpel that operates in the market for less than a week per month — waiting with extreme patience for one precise pattern, then acting decisively and retreating to the safety of cash within days"

Classification	Analysis Period	Asset Class
High-Precision Mean-Reversion Strategy	2003–2025	S&P 500